Part A: Multiple choice

(12 / marks)

Be sure that you have 6 different questions.
Complete the following table by putting the correct answer letter in the table

Q1) A class contains 4 mathematics students, 5 physics students, and 6 chemistry students. Suppose that 3 students are randomly selected, what is the probability that exactly 2 of the students chosen are mathematics?

a) 0.97 b) 0.145 c) 0.16 d) 0.297

Q2) If 4 dices are rolled, the probability that all show different faces is

a) 0.36 b)0.46 c)0.555 d)0.2778

Q3) Let X be a discrete random variable with $R_X = \{0,1,2\}$ and the P.D.F P(0)=0.2, P(1)=0.5 then E(3X)=

a) 2.2 b)1.83 c)2.4 d) 3.3

Q4) Three students are drawn from 8 students, of which 5 students have diabetes (مصاب بالسكري), what is the probability that at least one student has diabetes?

a) 0.098 b)0.98 c)0.82 d)0.792

Q5) if Cov (X,Y) = 3, E(Y) = 4 and E(1+2X) = 1.5 then E(XY) =

a) 4 b)2 c)3 d)1

Q6) The following table provides the joint probability distribution of X and Y

x \ y	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

The $P(X>1|Y=3)=$ a) 2/7 b) 2/5 c) 2/3 d) 2/9

Part B: Fill the Blanks

(8 marks)

Q7: In the test of computers, noticed that 80% of computers have no errors, 5% of computers have one error, and 20% of computers have two errors. Let x be the random variable representing the number of errors, then the mean of errors will be. ---------

Q8: Let A and B be two independent machines such that the probability that they both machine work is 1/6 and the probability that machine A works and machine B does not work is 1/3. Then the probability that machine B works is. ---------

Q9: Two independent firefighting components A, B, work such that p(A)=0.3, p(B)=0.6. Find the probability that both of them will fail. ---------

Q10: Let X be a continuous random variable with a Probability density function $f(x) = 3x^2 - a, \quad x \in [-1,3]$ *find* *a.*

Part C: Written Questions (10 / marks)

Question 11: Consider the following table that presents a random sample of 100 persons selected from the national Institute of diabetes. One person is selected at random from the sample

	male	female
Diabetic	38	14
Non-diabetic	22	26

what is the probability that the person is a male or the person has diabetes?

Question 12: A quadratic equation is given for three students S1 , S2 and S3, whose chances of solving it correctly are 0.3, 0.2, and 0.1, respectively. Find the probability that at least one student will solve the problem.

mathematics 2

A ; B-D-D B-A

7	8	9	10
0.45	$\frac{1}{3}$	0.82	6.75

(11) $P(M \cup D) = \frac{74}{100} =$

(12) $P(S_1 \cup S_2 \cup S_3) = 0.498$ // indep. events

indep. events

$= 0.865$ OR $1 - P(\overline{S_1 \cup S_2 \cup S_3})$